一、曲线积分

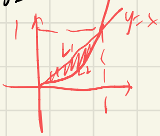
(1) $\oint_L (x^2+y^2)^n ds$，L为圆周 $x=a\cos t$，$y=a\sin t$，$0\le t\le 2\pi$

解：$\oint_L (x^2+y^2)^n ds = \int_0^{2\pi} (a^2)^n \sqrt{(a\sin t)^2+(a\cos t)^2}\, dt$

$= \int_0^{2\pi} a^{2n}\, a\, dt$

$= a^{2n+1}\, t \Big|_0^{2\pi} = 2\pi a^{2n+1}$

(2) $\oint_L x\, ds$　L: $y=x$ 与 $y=x^2$ 围成区域的整个边界



L_1 为 $x\in[0,1]$ 的 $y=x$

L_2 为 $x\in[0,1]$ 的 $y=x^2$

$\oint_L x\, ds = \int_0^1 x\sqrt{1+1^2}\, dx + \int_0^1 x\sqrt{1+(2x)^2}\, dx$

$= \sqrt{2}\int_0^1 x\, dx + \int_0^1 x\sqrt{1+4x^2}\, dx$

$= \sqrt{2}\, \frac{1}{2}x^2 \Big|_0^1 + \frac{1}{2\times 4}\int_0^1 \sqrt{1+4x^2}\, d(4x^2+1)$

$= \frac{\sqrt{2}}{2} + \frac{1}{8}\int_1^5 t^{\frac{1}{2}}\, dt$

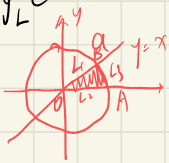
$= \frac{\sqrt{2}}{2} + \frac{1}{8}\, \frac{2}{3} t^{\frac{3}{2}} \Big|_1^5$

$= \frac{\sqrt{2}}{2} + \frac{1}{8}(5\sqrt{5}-1)\, \frac{2}{3}$

$= \frac{\sqrt{2}}{2} + \frac{1}{12}(5\sqrt{5}-1)$

$= \frac{5\sqrt{5}+6\sqrt{2}-1}{12}$

(3) $\oint_L e^{\sqrt{x^2+y^2}} ds$　L: $x^2+y^2=a^2$，$y=x$ 及x轴在第一象限所围成的扇形



$L = L_{OA} + L_{OB} + L_{AB}$

L_{OA}: $\int_{OA} e^{\sqrt{x^2+y^2}} ds = \int_{OA} e^x\, 1\, dx = \int_0^a e^x dx = e^a - 1$

L_{OB}: $(y=x)$　$\int_{OB} e^{\sqrt{2}x} \sqrt{2}\, dx = \int_0^{\frac{a}{\sqrt{2}}} e^{\sqrt{2}x} d(\sqrt{2}x) = e^a - 1$

L_{AB}: 设 $\begin{cases} x=a\cos t \\ y=a\sin t \end{cases}$　L_{AB}: $\int_{AB} e^a ds = \int_0^{\frac{\pi}{4}} e^a \sqrt{(a\sin t)^2+(a\cos t)^2}\, dt$

$= \int_0^{\frac{\pi}{4}} e^a\, a\, dt = a e^a \frac{\pi}{4}$

$\therefore \oint_L e^{\sqrt{x^2+y^2}} ds = 2e^a - 2 + a e^a \frac{\pi}{4}$

(4) $\int_\Gamma \frac{1}{x^2+y^2+z^2} ds$, Γ: $x=e^t\cos t$, $y=e^t\sin t$, $z=e^t$, $t\in[0,2]$

$$\int_\Gamma \frac{1}{x^2+y^2+z^2} ds = \int_0^2 \frac{1}{e^{2t}\cos^2 t+e^{2t}\sin^2 t+e^{2t}} \sqrt{(e^t\cos t-e^t\sin t)^2+(e^t\cos t+e^t\sin t)^2+(e^t)^2}\, dt$$

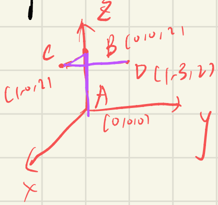
$$=\int_0^2 \frac{1}{2e^{2t}} e^t \sqrt{1+(\cos^2 t+\sin^2 t)\times 2}\, dt$$

$$=\frac{\sqrt{3}}{2}\int_0^2 \frac{1}{e^t} dt = \frac{-\sqrt{3}}{2}\int_0^2 e^{-t} d(-t) = \frac{\sqrt{3}}{2}\int_{-2}^0 e^t dt = e^t\Big|_{-2}^0 \times \frac{\sqrt{3}}{2}$$

$$=\frac{\sqrt{3}}{2}(e^0-e^{-2}) = \frac{\sqrt{3}}{2}(1-e^{-2})$$

(5) $\int_\Gamma x^2yz\, ds$, Γ: ABCD , A(0,0,0), B(0,0,2) , C(1,0,2) D(1,3,2)

解:



$\int_{AB}$ $x=y=0$, $\int_{AB} x^2yz\, ds=0$

$\int_{BC}$ $y=0$, $\int_{BC} x^2yz\, ds=0$

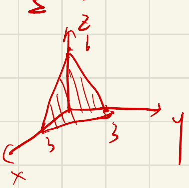
$\int_{CD}$ $x=1, z=2$, $y\in[0,3]$

$$\int_\Gamma x^2yz\, ds = \int_\Gamma 2y\, dy = 2\int_0^3 y\, dy = 2\times \frac{1}{2}y^2\Big|_0^3 = 9$$

$$\Rightarrow \int_\Gamma x^2yz\, ds = 9.$$

二. 曲面积分

(1) $\iint_\Sigma (2xy-2x^2-x+z)\, dS$. Σ为 $2x+2y+z=6$ 在第一象限中的部分.



$z=6-2x-2y$

$dS=\sqrt{1^2+2^2+2^2}\, dxdy = 3\,dxdy$

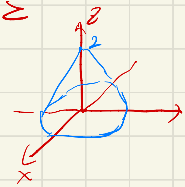
$\Rightarrow x+y=3$, $y=3-x$

$$\iint_\Sigma (2xy-2x^2-x+z)\, dS = 3\iint_{D_{xy}} (2xy-2x^2-x+6-2x-2y)\, dxdy$$

$$=3\int_0^3 dx\int_0^{3-x} (6-3x-2y-2x^2+2xy)\, dy = 3\int_0^3 (3x^3-10x^2+9)\, dx$$

$$=-\frac{27}{4}$$

(2) $\iint\limits_{\Sigma} 3z\,ds$, $\Sigma: z=2-(x^2+y^2)$ 在xoy面上方的部分



$0\le x^2+y^2\le 2$

$$ds=\sqrt{1+z_x'^2+z_y'^2}\,dxdy$$
$$=\sqrt{1+4x^2+4y^2}\,dxdy$$

$$\iint\limits_{\Sigma} 6-3(x^2+y^2)\,ds=\iint\limits_{\Sigma}[6-3(x^2+y^2)]\sqrt{1+4x^2+4y^2}\,dxdy$$
$$=6\iint\limits_{\Sigma}\sqrt{1+4x^2+4y^2}\,dxdy-3\iint\limits_{\Sigma}(x^2+y^2)\sqrt{1+4x^2+4y^2}\,dxdy$$
$$=6\int_0^{2\pi}d\theta\int_0^{\sqrt2}\sqrt{1+4r^2}\,r\,dr-3\int_0^{2\pi}d\theta\int_0^{\sqrt2}r^2\sqrt{1+4r^2}\,r\,dr$$
$$=\frac{12\pi}{8}\int_0^{\sqrt2}\sqrt{1+4r^2}\,d(4r^2+1)-6\pi\int_0^{\sqrt2}r^3\sqrt{1+4r^2}\,dr$$
$$=\frac32\pi\,\frac23(4r^2+1)^{\frac32}\Big|_0^{\sqrt2}-\frac{6\pi}{2}\int_0^{\sqrt2}r^2\sqrt{1+4r^2}\,d(r^2)$$
$$=\pi(9^{\frac32}-1^{\frac32})-3\pi\int_0^2 x\sqrt{1+4x}\,dx$$
$$=26\pi-\frac32\pi\int_0^2\sqrt{1+4x}\,d(x^2)$$
$$=26\pi-\frac32\pi\,x^2\sqrt{1+4x}\Big|_0^2+\frac32\pi\int_0^2 x^2\,d\sqrt{1+4x}$$
$$=26\pi-\frac32\pi\times4\times3+\frac32\pi\int_0^2\frac{1\times4}{2\sqrt{1+4x}}x^2\,dx$$
$$=8\pi+\frac32\pi\int_0^2\frac{2x^2}{\sqrt{1+4x}}\,dx \qquad (\sqrt{1+4x}=t.)$$
$$=8\pi+\frac32\pi\int_1^3\frac{\frac{(t^2-1)^2}{8}}{t}\,d\left(\frac{t^2-1}{4}\right) \qquad x=\frac{t^2-1}{4}$$
$$=8\pi+\frac32\pi\int_1^3\frac{(t^2-1)^2}{8t}\,\frac{t}{2}\,dt$$
$$=8\pi+\frac{3}{2\times16}\pi\int_1^3(t^2-1)^2\,dt$$
$$=8\pi+\frac{3}{32}\pi\left(\frac15t^5-\frac23t^3+t\right)\Big|_1^3$$
$$=8\pi+\frac{3}{32}\pi\left[\frac{243}{5}-18+3-\frac15+\frac23-1\right]=\frac{111}{10}\pi$$

(3) $\iint\limits_{\Sigma}(x^2+y^2)\,dS$, Σ: $z^2=3(x^2+y^2)$, $z=0$, $z=3$

$$x^2+y^2\in[0,\sqrt{3}]$$

$$dS=\sqrt{1+z_x^2+z_y^2}\,dxdy$$

$$=\sqrt{1+\frac{9x^2}{3(x^2+y^2)}+\frac{9y^2}{3x^2+3y^2}}\,dxdy=2dxdy$$

$$\iint\limits_{\Sigma}(x^2+y^2)\,dS=\iint\limits_{D_{xy}}2(x^2+y^2)\,dxdy$$

$$=2\int_0^{2\pi}d\theta\int_0^{\sqrt{3}}r^2\cdot r\,dr=4\pi\ \frac{1}{4}r^4\Big|_0^{\sqrt{3}}$$

$$=4\pi\times\frac{1}{4}3^2=9\pi.$$

(4) $\iint\limits_{\Sigma}(x+y+z)\,dS$ $\quad\Sigma$: $x^2+y^2+z^2=a^2$, $z\geq h$, $(0<h<a)$

$$z=\sqrt{a^2-x^2-y^2}\ ,\quad x^2+y^2\in[0,\ a^2-h^2]$$

$\iint\limits_{\Sigma}(x+y)\,dS=0$ (对称性)

原 $=\iint\limits_{\Sigma}z\,dS=\iint\limits_{D_{xy}}\sqrt{a^2-x^2-y^2}\sqrt{1+\frac{x^2}{a^2-x^2-y^2}+\frac{y^2}{a^2-x^2-y^2}}\,dxdy$

$$=\iint\limits_{D_{xy}}\sqrt{a^2-x^2-y^2}\ \frac{a}{\sqrt{a^2-x^2-y^2}}\,dxdy$$

$$=\iint\limits_{D_{xy}}a\,dxdy=a\iint\limits_{D_{xy}}1\,dxdy=a\pi(a^2-h^2).$$

(5) $\iint\limits_{\Sigma}(xy+yz+zx)\,dS$

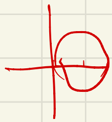
$\Sigma: z=\sqrt{x^2+y^2},\ x^2+y^2=2ax.$

$z=\sqrt{x^2+y^2}$

$dS=\sqrt{1+\left(\frac{x}{\sqrt{x^2+y^2}}\right)^2+\left(\frac{y}{\sqrt{x^2+y^2}}\right)^2}\,dxdy=\sqrt{2}\,dxdy$

解 $=\sqrt{2}\iint\limits_{D_{xy}} xy+(x+y)\sqrt{x^2+y^2}\,dxdy$

$x^2+y^2=2ax \qquad (x-a)^2+y^2=a^2$



关于y轴对称

解 $=\sqrt{2}\iint\limits_{D_{xy}} x\sqrt{x^2+y^2}\,dxdy$

$=\sqrt{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}d\theta\int_0^{2a\cos\theta} r^2\cos\theta\, r\,dr$

$=8\sqrt{2}a^4\int_0^{\frac{\pi}{2}}\cos^5\theta\,d\theta=8\sqrt{2}a^4\left(\frac{4}{5}\times\frac{2}{3}\right)=\frac{64}{15}\sqrt{2}a^4.$

三. 第二型曲线积分

(1) $\int_L(x^2-2xy)dx+(y^2-2xy)dy$, $L: y=x^2$ 从 $(-1,1)$ 到 $(1,1)$

$x\in[-1,1]$

解 $=\int_L(x^2-2xy)dx+\int_L(y^2-2xy)dy$

$=\int_{-1}^{1}(x^2-2x\,x^2)dx+\int_{-1}^{1}(x^4-2x^3)\,2x\,dx$

$=\int_{-1}^{1}\left[x^2-2x^3+(x^4-2x^3)2x\right]dx$

$=\int_{-1}^{1}(2x^5-4x^4-2x^3+x^2)dx$

$=\frac{1}{3}x^6-\frac{4}{5}x^5-\frac{1}{2}x^4+\frac{1}{3}x^3\Big|_{-1}^{1}=-\frac{8}{5}+\frac{2}{3}=-\frac{14}{15}.$

(2) $\oint_L \frac{(x+y)dx-(x-y)dy}{x^2+y^2}$, $L: x^2+y^2=a^2$，逆时针方向.

$$\begin{cases} x=a\cos t \\ y=a\sin t \end{cases} \quad t: 0\to 2\pi$$

$$\oint_L \frac{(x+y)dx-(x-y)dy}{x^2+y^2}$$

$$= \frac{1}{a^2}\int_0^{2\pi}\left[(a\sin t+a\cos t)(-a\sin t)-(a\cos t-a\sin t)a\cos t\right]dt$$

$$= \frac{1}{a^2}\int_0^{2\pi}\left[-a^2\sin^2 t-a^2\cos t\sin t-a^2\cos^2 t+a^2\cos t\sin t\right]dt$$

$$= \frac{1}{a^2}\int_0^{2\pi}(-a^2)\,dt = \frac{1}{a^2}(-a^2)\,2\pi = -2\pi$$

(3) $\int_\Gamma x^2dx+z\,dy-y\,dz$　$\Gamma: x=k\theta, y=a\cos\theta, z=a\sin\theta$，$\theta\in[0,\pi]$

$$原式 = \int_0^\pi k^2\theta^2 k\,d\theta + \int_0^\pi a\sin\theta\, a(-\sin\theta)\,d\theta - \int_0^\pi a\cos\theta\, a\cos\theta\,d\theta$$

$$= \int_0^\pi k^3\theta^2\,d\theta - \int_0^\pi a^2(\sin^2\theta+\cos^2\theta)\,d\theta$$

$$= k^3\frac{1}{3}\theta^3\Big|_0^\pi - \int_0^\pi a^2\,d\theta$$

$$= k^3\frac{1}{3}\pi^3 - a^2\pi = \frac{k^3\pi^3}{3} - \pi a^2$$

(4) $\int_\Gamma x\,dx+y\,dy+(x+y-1)\,dz$，$\Gamma\ (1,1,1)\to(2,3,4)$

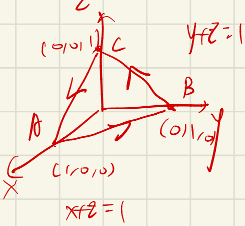
$$\begin{cases} x=1+k \\ y=1+2k \\ z=1+3k \end{cases} \quad k: 0\to 1$$

$$原式 = \int_0^1 (1+k)\,dk + \int_0^1 (1+2k)\times 2\,dk + \int_0^1 (2+3k-1)\times 3\,dk$$

$$= \int_0^1 (1+k+2+4k+3+9k)\,dk = \int_0^1 (6+14k)\,dk$$

$$= 6k+7k^2\Big|_0^1 = 13.$$

(5) $\oint_\Gamma dx-dy+y\,dz$, Γ: ABCA. A(1,0,0), B(0,1,0), C(0,0,1))



$$\oint_{AB} dx-dy+y\,dz = \int_1^0 dx-(-1)\,dx = -2$$

$$\oint_{BC} dx-dy+y\,dz = \int_0^1 y\,dz + \int_0^1 dz.$$

$$= \int_0^1 (2-z)\,dz = 2z-\frac{1}{2}z^2 \Big|_0^1 = \frac{3}{2}$$

$$\oint_{CA} dx-dy+y\,dz = \int_0^1 dx + \int_0^1 y(-1)\,dx = 1$$

$$\therefore \oint_\Gamma dx-dy+y\,dz = -2+\frac{3}{2}+1 = \frac{1}{2}.$$